It is nevertheless overwhelmingly tempting to wonder whether time, like space, might be emergent rather than fundamental, and whether entanglement might have anything to do with it. The answer is yes on both counts, although the details remain a little sketchy.

If we take the Schrödinger equation at face value, time seems to be right there in a fundamental way. Indeed, it immediately follows that the universe lasts eternally toward both the past and future, for almost all quantum states. You might think that this conflicts with the oft-repeated fact that the Big Bang was the beginning of our universe, but we don't actually know that oft-repeated fact to be true. That's a prediction of classical general relativity, not of quantum gravity. If quantum gravity operates according to some version of the Schrödinger equation, then for almost all quantum states, time runs from minus infinity in the past to plus infinity in the future. The Big Bang might be simply a transitional phase, with an infinitely old universe preceding it.

We have to say "almost all" in these statements because there is one loophole. The Schrödinger equation says that the rate of change of the wave function is driven by how much energy the quantum system has. What if we consider systems whose energy is precisely zero? Then all the equation says is that the system doesn't evolve at all; time has disappeared from the story.

You might think it's extremely implausible that the universe has exactly zero energy, but general relativity suggests you shouldn't be so sure. Of course there seem to be energy-containing things all around us—stars, planets, interstellar radiation, dark matter, dark energy, and so on. But when you go through the maths, there is also a contribution to the energy of the universe from the gravitational field itself, which is generally negative. In a closed universe—one that wraps around on itself to form a compact geometry, like a three-dimensional sphere or torus, rather than stretching to infinity—that gravitational energy precisely cancels the positive energy from everything else. A closed universe has exactly zero energy, regardless of what's inside.

That's a classical statement, but there's a quantum-mechanical analogue that was developed by John Wheeler and Bryce DeWitt. The Wheeler-DeWitt equation simply says that the quantum state of the universe doesn't evolve at all as a function of time.

This seems crazy, or at least in flagrant contradiction to our observational experience. The universe certainly seems to evolve. This